Some more C++ Assignments

1. Write a C++ program to calculate Simple and Compound interest.

2. Write a C++ Program to transpose one matrix and display both the matrices with proper prompt.

3. Write a C++ program to process students' marks with the help of classes. The class has private variables for name, mark 1, mark 2, mark 3. It has two member functions – getdata( ) – to get input and result( ) – to print the results. All subjects mark must be >= 50 for pass otherwise fail. Implement this program for n number of students.

4. Write a C++ program that allows the user to input 5 grades, i.e. marks between 0 - 100. The program must calculate the average mark, and state the number of marks less than 65.

5. Write a menu driven C++ program to add, subtract and multiply given two matrices defined in class. Modify your program using operator overloading method.

6. Create a class DONOR that contains donor number, donor name, age address, sex and blood group. Write a Menu driven C++ program to display the number, name and address of the donors for the following categories:
   a. blood donors having the blood group O+
   b. blood donors in the age group between 18 to 25
   c. female donors having blood group A in the age between 21 and 24.

7. An electricity board charges the following rates to domestic users:

   For the first 100 units – ₹. 5.5P per unit
   For next 200 units – ₹. 7.25P per unit
   Beyond 300 units – ₹. 8.75P per unit

   All users are charged a minimum of ₹. 60.

   If the total amount is more than ₹. 750 then an additional surcharge of 10% is added on the higher slab units.

   Write a C++ program to read the names of users and number of units consumed and print out the charges with names.

8. Implement the calculator through a C++ program (use switch). The program should ask the user to input two operands and an operator. Your program should print the result of performing the required operation on the input operands as the following output:

   Enter operand 1: 5
   Enter operand 2: 10
   Enter operation: /

   The result of 5/10 is 0.5

9. Create Time Class which contains the members as hour, minute and second. Write a C++ program that accepts member data for one Time class object from the keyboard. Then, ask the user to perform one of the following functions to that Time object:
    - AddHours
    - AddMinutes
    - AddSeconds

And finally display the newly modified Time information.

10. Write a menu driven program to perform the following options using separate functions. Accept a number and then, do the following operations on this number:
    1. Prime number or not
    2. Odd or Even
    3. Positive or Negative
    4. Exit

11. Write a function reverse() that reverses the character strings. Use it to write a program that reverses its input a line at a time.

12. Write a C++ Program to Compare Two given Strings and display the result.

13. Write a program in C++ to determine how many of the characters are vowels and how many are consonants in a given line of text. Also terminate the string when the input character encountered is other than the alphabets (a-z or A-Z) and Blank spaces.
    Hint:
    a. When the input string is 'C plus plus FOR BEGINNERS, TEST YOUR C plus plus PROGRAMMING STRENGTHS'. Program will consider the string 'C plus plus FOR BEGINNERS' only because ',' is encountered.
    b. When the input string is 'C++ LEARINING IS NOT A PROBLEM'. Program will consider the character 'C' only because the '+' is encountered.

14. Create a C++ class for a stock item abstract data type. It should have the attributes of stock levels (an integer) and unit price (a float). Define the methods to return the values of these two attributes and to set them using parameters. Add two more methods to allow stock receipt and issues updating the stocklevel as appropriate. Write a menu driven C++ program to solve the problem.

15. Raising a number n to power p is the same as multiplying by itself t times. Write a function called power( ) that makes a double value for n and an int value for p and returns the result as double value. Use default argument of 2 for p, so that if this argument is omitted, the number will be squared. Write a main( ) function that gets values from the user to test this function.